

Fashion Merchandising Dashboard — Full Build Specification

Overview

Build a comprehensive fashion merchandising analytics dashboard for a multi-brand retail business operating across physical stores in Kenya. The business sells women's apparel and accessories under the brands **Vivo**, **Safari**, and **Zoya**, with approximately 500 active styles across 30 stores. The dashboard is a multi-page tool used daily by merchandisers, buyers, planners, and executives to monitor portfolio health, manage inventory, and make restocking and markdown decisions.

The tool must be data-driven, action-oriented, and role-aware. Every metric should lead to a recommended action. The visual style should be clean and professional: white card backgrounds on a light grey page background, a blue/teal/green/amber/red colour system for status and trends, consistent typography hierarchy, and interactive drill-down from summary to detail.

Design System

Colour palette:

- Background: #F5F6FA (light grey page)
- Card background: #FFFFFF
- Primary blue: #2D6BE4 (main data, Vivo brand)
- Teal: #4ECDC4 (secondary data, Safari brand)
- Purple: #7B5EA7 (tertiary, Zoya brand)
- Green: #27AE60 (healthy / positive / on track)
- Amber: #F5A623 (warning / monitor / at risk)
- Red: #E84040 (critical / negative / overdue)
- Text primary: #1A2B4A
- Text secondary: #8A94A6
- Border / divider: #E8EAF0

Typography: Single sans-serif font family. Hero metric = 20–24px bold. Section title = 10–11px bold. Card label = 8–9px medium. Data label = 7–8px regular.

Card layout: Every metric or chart lives inside a white rounded card with subtle shadow. KPI cards have a 4px coloured left accent bar matching their status colour. All cards include a title, primary value, and a secondary context line (e.g. "vs subcategory average" or "last 6 months").

Status colour logic:

- Green = healthy, on track, WOC 2–8 weeks, SOR > 65%, Full Price % > 85%
- Amber = monitor, at risk, WOC 8–20 weeks or < 2 weeks, SOR 40–65%
- Red = critical, overdue, WOC < 1 week or out of stock, SOR < 40%
- Teal = overstocked, WOC > 20 weeks

Interactive behaviour: Every chart and KPI card should be clickable to drill down to the next level of detail. Filters (brand, subcategory, tier, status, date range) should persist across all pages. A global search bar should allow jumping directly to any style by name or style number.

Data Model

The core dataset is a style-level table with one row per active style. Key fields include:

Field	Description
Style	Style name
Style Number	Unique SKU identifier
Brand	Vivo / Safari / Zoya / Safari by Vivo
Subcategory	Product category (e.g. Maxi Dresses, Full Length Pants)
Tier	Tier 1 (core) / Tier 2 / Tier 3 / Tier 4 (new/trial)
Launch Date	Date first sold
Age (weeks)	Weeks since launch
Reorder Count	Number of times restocked since launch
Current Stock	Total units currently in stock across all stores
WOC	Weeks of Cover = Current Stock / Weekly Avg
Weekly Avg	Average weekly units sold
Last Sale (days)	Days since last unit was sold

Full Price %	Percentage of units sold at full price
Full Price (KES)	Full retail price
Avg Price (KES)	Average achieved selling price
Cost Price (KES)	Wholesale / landed cost (used for GM calculation)
Units Since Launch	Cumulative units sold lifetime
Revenue Since Launch (KES)	Cumulative revenue lifetime
Units (6m)	Units sold in last 6 months
Revenue (6m) (KES)	Revenue in last 6 months
SOR Since Launch %	Sell-Out Rate lifetime
SOR (6m) %	Sell-Out Rate last 6 months
Status	On Track / At Risk / Overdue
Recommended Action	Text description of suggested action

Extended data (store level): For each style, a second table holds per-store data: Store Name, Region, Store Tier (Flagship / Standard / Boutique / Regional), Current Stock at Store, Units Sold (6m) at Store, Revenue (6m) at Store, Rate of Sale (units/week) at Store, Weeks of Cover at Store.

Derived fields to calculate:

- Gross Margin % = $(\text{Avg Price} - \text{Cost Price}) / \text{Avg Price} \times 100$
- Gross Margin KES = $(\text{Avg Price} - \text{Cost Price}) \times \text{Units Sold}$
- Subcategory Percentile Rank = this style's position within its subcategory for each metric
- Transfer Recommendation = flag stores with WOC > 3 weeks as donors, stores with WOC < 1.5 weeks or OOS as recipients

Page Structure — 12 Pages Total

Page 1 — Executive Overview

Purpose: The landing page. Gives executives and directors a 60-second read of total portfolio health across all brands, subcategories, and stores. Every metric links to a deeper page.

Audience: CEO, Commercial Director, Head of Merchandising.

KPI cards (row 1):

- Total Active Styles (count, with On Track / At Risk / Overdue split)
- Total Current Stock Units (all stores combined)
- Revenue Last 6 Months (KES, with avg per style)
- Units Sold Last 6 Months (with weekly velocity)
- Average Full Price % (portfolio-wide)
- Average SOR Last 6 Months %

Charts (rows 2–3):

- Horizontal bar chart: Revenue by Brand (6m), colour-coded by brand
- Horizontal bar chart: Top 5 Subcategories by Revenue (6m)
- Donut chart: Revenue share by Tier (Tier 1 / 2 / 3 / 4)
- Bar chart: Style Status breakdown (On Track / At Risk / Overdue) with counts
- Bar chart: Weeks of Cover distribution (0–4 / 4–8 / 8–12 / 12–20 / 20+ weeks buckets)
- Bar chart: Full Price % distribution (<70% / 70–85% / 85–95% / >95% buckets)
- Action summary panel: Four colour-coded rows showing count of styles in each recommended action category (Healthy / On Review / Marketing Push Needed / Overdue)

Filters: Brand, Subcategory, Tier, Status, Date Range, Store.

Page 2 — Sales Performance

Purpose: Deep dive into revenue and units. Identifies which styles, subcategories, and brands are driving the business and where velocity is strong or weak.

Audience: Head of Merchandising, Buyers, Sales Managers.

KPI cards:

- Revenue (6m) total
- Units Sold (6m) total
- Average Unit Retail (AUR) — avg selling price across all styles

- Average Full Price % across portfolio
- Total Weekly Velocity (units/week across all styles)
- Number of styles with >95% Full Price (premium performers)

Charts:

- Horizontal bar chart: Top 10 revenue-generating styles (6m), colour intensity by revenue tier
 - Vertical bar chart: Revenue by Subcategory (6m), all subcategories
 - Grouped bar chart: Units sold by Brand (6m)
 - Grouped bar chart: Average Full Price vs Average Selling Price by Tier
 - Donut chart: Full Price % mix (<70% / 70–85% / 85–95% / >95%)
-

Page 3 — Inventory & Stock Health

Purpose: Monitor stock levels across the portfolio. Identify overstocked and understocked positions before they become problems.

Audience: Planners, Allocators, Supply Chain team.

KPI cards:

- Total Stock Units (all stores)
- Average WOC across portfolio
- Number of styles with WOC > 20 weeks (overstock risk)
- Number of styles with zero stock (stockout)
- Number of styles with no sale in last 30 days
- Number of styles with WOC < 4 weeks (reorder urgency)

Charts:

- Horizontal bar chart: Top 10 styles by current stock, colour-coded by WOC risk
 - Horizontal bar chart: Average WOC by subcategory (highest-risk subcategories)
 - Bar chart: Current stock by Brand
 - Bar chart: Styles by last-sale recency (sold today / 1–7 days / 8–30 days / 31–60 days / 60+ days)
 - Donut chart: Stock distribution by Tier
 - Stacked bar chart: WOC bucket breakdown by subcategory
-

Page 4 — Sell-Through & Markdown Analysis

Purpose: Track how efficiently inventory is converting to sales. Identify styles at risk of requiring deep markdowns before it is too late.

Audience: Merchandisers, Buyers, Finance.

KPI cards:

- Average Lifetime SOR % (portfolio)
- Average SOR (6m) % (portfolio)
- Average Full Price % (portfolio)
- Number of styles with SOR (6m) > 90% (star performers)
- Number of styles with SOR (6m) < 40% (markdown risk)
- Number of styles with Full Price % < 70% (already heavily discounted)

Charts:

- Horizontal bar chart: SOR (6m) by subcategory, colour-coded against 60% target line
 - Scatter plot: Lifetime SOR % vs 6m SOR % (each dot = one style, colour = SOR band)
 - Grouped bar chart: Full Price % vs SOR (6m) % by Brand
 - Bar chart: SOR (6m) distribution buckets (<40% / 40–60% / 60–80% / 80–100%)
 - Grouped bar chart: Full Price % vs SOR (6m) % by Tier
-

Page 5 — Category Performance

Purpose: Strategic view of how each subcategory is performing. Used for range planning and investment decisions.

Audience: Head of Merchandising, Buyers, Range Planners.

KPI cards:

- Top subcategory by revenue (6m)
- Highest SOR subcategory (6m)
- Subcategory with most active styles
- Subcategory with highest average Full Price %
- Subcategory with highest total weekly velocity

Charts:

- Bubble chart: Revenue (6m) on x-axis, SOR (6m) on y-axis, bubble size = current stock. Each bubble = one subcategory. Colour = SOR health (green/amber/red). This is the

centrepiece chart — it shows which categories are both high-revenue and high-sell-through vs those that are revenue-heavy but slow-moving.

- Horizontal bar chart: Active style count by subcategory
 - Horizontal bar chart: Total weekly velocity by subcategory
 - Horizontal bar chart: Average Full Price % by subcategory, with target line at 85%
 - Donut chart: Revenue share — top 4 subcategories vs rest
-

Page 6 — Style Lifecycle & Age Analysis

Purpose: Understand the age profile of the portfolio. Identify long-running core styles vs new launches vs styles that have aged out.

Audience: Buyers, Range Planners, Head of Merchandising.

KPI cards:

- Average style age (weeks)
- Number of styles launched in current year
- Number of styles launched in prior year
- Average reorder count across portfolio
- Maximum reorder count (the most-reordered style)
- Number of styles never reordered (Tier 4 trial styles)

Charts:

- Bar chart: Styles launched by year (2019–2026), colour-coded by era
 - Bar chart: Portfolio age distribution (0–26 weeks / 27–52 / 53–104 / 105–208 / 208+ weeks)
 - Donut chart: Reorder count distribution (0× / 1–5× / 6–10× / 11–20× / 21+×)
 - Bar chart: Average reorder count by Tier (shows Tier 1 dominance)
 - Scatter plot: Style age (weeks) vs Lifetime SOR %, colour-coded by Tier
 - Table: Tier 1 top performers — Style name, Age, Reorder Count, Lifetime SOR
-

Page 7 — At-Risk Styles & Action Required

Purpose: The action dashboard. Surfaces every style that needs immediate attention, ranked by urgency and stock exposure.

Audience: Merchandisers, Buyers, Store Operations.

KPI cards:

- Total At Risk styles (count + % of portfolio)
- Total Overdue styles (immediate action required)
- Total On Review styles
- Estimated stock value at risk (units × avg price for at-risk styles)
- Total Healthy styles (count + % of portfolio)

Charts:

- Horizontal bar chart: Top 10 at-risk styles by current stock exposure, colour = days since last sale
 - Donut chart: Portfolio action status (Healthy / On Review / Marketing Push / Overdue)
 - Horizontal bar chart: At-risk style count by subcategory
 - Grouped bar chart: Status breakdown (Healthy / On Review / At Risk) by age group
 - Action priority panel: Four colour-coded action cards with style counts and urgency descriptions
-

Page 8 — Replenishment & Reorder Planning

Purpose: Identify which styles need to be reordered now, soon, or can wait. Prioritise purchase orders by velocity and WOC.

Audience: Planners, Allocators, Supply Chain.

KPI cards:

- Total weekly velocity across portfolio (units/week)
- Number of styles with WOC < 4 weeks (urgent reorder)
- Number of styles with WOC 4–8 weeks (reorder soon)
- Average reorder count per style
- Number of styles currently out of stock

Charts:

- Horizontal bar chart: High-velocity styles ranked by current WOC, colour = urgency. Labels show current stock and WOC. Vertical lines at 4-week and 8-week trigger points.
- Horizontal bar chart: Average WOC by subcategory, with 12-week target line
- Bar chart: Weekly velocity distribution (0–5 / 5–20 / 20–50 / 50–100 / 100+ units/week)
- Dual-axis bar + line chart: Average reorder count (bars) and style count (line) by Tier

- Donut chart: WOC bucket mix across portfolio
-

Page 9 — Financial Performance

Purpose: Revenue, pricing, and gross margin analysis. Used for financial reporting and margin management.

Audience: Finance, Commercial Director, CEO.

KPI cards:

- Total Revenue (6m) KES
- Estimated Lifetime Revenue KES
- Average Full Price KES
- Average Selling Price KES
- Average Price Realisation % (avg selling / full price)
- Estimated Gross Margin (6m) KES and %

Charts:

- Vertical bar chart: Revenue by subcategory (6m), all subcategories
 - Grouped bar + line chart: Full price, avg selling price (bars) and price realisation % (line) by Brand
 - Bar chart: Average revenue per style by Tier (shows Tier 1 ROI)
 - Grouped bar chart: Lifetime revenue vs 6m revenue by Brand
 - Donut chart: Full price distribution (<KES 2K / 2K–4K / 4K–6K / 6K–8K / >8K)
-

Page 10 — New Arrivals & Pipeline

Purpose: Track the performance of recently launched styles. Identify early winners to back and early underperformers to address before stock builds.

Audience: Buyers, Range Planners, Merchandisers.

KPI cards:

- Styles launched in current year (count)
- Styles launched in prior year (count)
- Average age of current-year launches (weeks)
- Estimated revenue from current-year launches (6m)
- % of current-year launches On Track

Charts:

- Horizontal bar chart: Current-year launches by subcategory
 - Line chart: SOR ramp curves by Tier (cumulative SOR % by week since launch). Shows how quickly each tier reaches the 60% target.
 - Donut chart: Status of current-year launches (On Track / At Risk / Overdue)
 - Horizontal bar chart: Top recent launches by SOR %, with age in weeks labelled
 - Stacked bar chart: Monthly launch cadence (current year), stacked by On Track vs At Risk
 - Donut chart: Current-year launches by Brand
-

Page 11 — Style Deep Dive (Page 1 of 2)

Purpose: A complete single-style analysis page. Accessed by clicking any style name anywhere in the tool. Shows everything known about one style — its history, performance vs peers, financial contribution, and recommended actions.

Audience: Any user who needs to investigate a specific style.

Header: Style name (large, bold) + Style Number • Brand • Subcategory • Tier • Launch Date • As at [date]. Status badge (On Track / At Risk / Overdue) top right. Recommendation summary (e.g. "RESTOCK URGENTLY") top right.

KPI cards (6 cards):

- Revenue (6m) — with delta vs subcategory average
- Units Sold (6m) — with delta vs subcategory average
- SOR (6m) % — with delta vs subcategory average
- Gross Margin % — with KES per unit and cost price shown
- Weeks of Cover — with delta vs subcategory average, red if critical
- Reorder Count — with style age in weeks

Charts:

- Line chart with fill: Weekly sales trend, trailing 52 weeks. Includes current average velocity line, subcategory average benchmark line, and vertical markers at each reorder event (labelled RO1, RO2 etc.).
- Percentile bar chart: This style's percentile rank within its subcategory on 5 metrics — Revenue (6m), Units (6m), SOR (6m), Weekly Velocity, Full Price %. Bars show percentile height (0–100), with a median line at 50th. Colour = green if >75th, amber if 25–75th, red if <25th.

- Gross Margin waterfall: Full Price → Discount → Avg Selling Price → Cost Price → Gross Margin. Each step labelled with KES value.
 - Monthly revenue bar chart: Last 12 months, with line overlay showing trend.
 - Style lifecycle timeline: Horizontal timeline from launch to today (in weeks). Marks every reorder event. Marks key milestones (launch, Tier promotion, today). Overlays cumulative units sold curve on secondary axis.
 - AI Recommendation Engine panel: 4 colour-coded action cards — one for restocking decision, one for retirement decision, one for inter-store transfer recommendation, one for margin/pricing opportunity. Each card has a coloured left border, bold title, and 2-line explanation.
-

Page 12 — Style Deep Dive (Page 2 of 2): Store Performance Detail

Purpose: The store-level breakdown for one style. Shows exactly how the style is performing in each of the 30 stores — revenue, units, rate of sale, stock, weeks of cover — and makes specific transfer recommendations.

Audience: Planners, Allocators, Store Operations, Regional Managers.

Header: Same style name as Page 11 + " — Store Performance Detail" • "Page 2 of 2". Total 6m revenue and total 6m units shown top right.

KPI cards (6 cards):

- Total Stock across all stores
- Number of stores out of stock
- Number of stores with WOC < 2 weeks
- Best performing store by revenue
- Average Rate of Sale per store (units/week)
- Estimated Gross Margin for 6m period (KES)

Charts:

- Bar chart: Revenue per store (6m), all 30 stores sorted by revenue descending. Colour-coded by WOC risk (red/amber/green/teal). Background shading bands group stores by tier (Flagship / Standard / Boutique / Regional). Top 10 stores labelled with revenue value.
- Bar chart: Rate of Sale per store (units/week), same sort order and colour coding. Portfolio average ROS shown as a dashed benchmark line.
- Bar chart: Units sold per store (6m), same sort order.

- Bar chart: Weeks of Cover per store, sorted high to low (overstocked on left, OOS on right). Dashed lines at 1-week (critical) and 2-week (reorder trigger) thresholds. Each bar labelled with WOC value or "OOS".

Full Store Data Table: One row per store, all 30 stores. Columns: Store Name, Tier, Revenue 6m (KES), Units 6m, ROS /week, Current Stock, WOC, Gross Margin Contribution (KES), Action. The Action column is colour-coded: RESTOCK (red) / REORDER SOON (amber) / TRANSFER OUT (teal) / MONITOR (blue) / HEALTHY (green). Alternating row shading for readability.

Transfer Plan panel: Two sections separated by a divider.

- Top section (DONOR STORES): Lists stores with WOC > 3 weeks and stock > 1 unit. Shows store name, ROS, units available to give, and current WOC.
- Bottom section (RECIPIENT STORES): Lists stores with WOC < 1.5 weeks or OOS and ROS ≥ 1.5 units/week. Shows store name, ROS, units to receive, and current WOC.
- Footer note: Total transferable units and remaining reorder quantity needed.

Regional Summary donut: Revenue share by geographic region (Nairobi / Rift Valley / Western / Coast / Central / Eastern), with percentage labels. Summary text below showing Nairobi share vs regional store share.

Navigation Structure

The tool uses a left sidebar with icon-based navigation. Each icon corresponds to one of the 12 pages. The sidebar also shows active filters as removable tags. A global date range selector and "Compare to: Previous Period" toggle sit in the top bar on every page.

Clicking any style name anywhere in the tool navigates to Page 11 (Style Deep Dive Page 1). A "View Store Detail" button on Page 11 navigates to Page 12. Breadcrumb navigation shows the path back to the summary page.

Filter System

Global filters available on every page:

- **Brand:** All / Vivo / Safari / Zoya / Safari by Vivo
- **Subcategory:** Multi-select dropdown of all subcategories
- **Tier:** All / Tier 1 / Tier 2 / Tier 3 / Tier 4
- **Status:** All / On Track / At Risk / Overdue

- **Date Range:** Custom date picker with presets (Last 4 weeks / Last 13 weeks / Last 26 weeks / Last 52 weeks / Season to Date / Year to Date)
- **Compare to:** Previous Period / Previous Year / None
- **Store / Region:** All / specific store or region
- **Price Range:** All / Under KES 2,000 / KES 2,000–5,000 / Over KES 5,000

Active filters are shown as removable tags at the top of the filter panel. A "Save View" button allows users to bookmark a filtered configuration.

Recommended Action Logic

The system should automatically assign a recommended action to every style based on the following rules:

Condition	Action	Colour
SOR (6m) > 65% AND WOC 2–12 weeks AND Full Price % > 80%	Healthy — maintain replenishment	Green
WOC < 2 weeks AND Weekly Avg > 5	Restock urgently — raise PO	Red
WOC = 0 (out of stock)	Out of stock — restock immediately	Red
Last Sale > 60 days AND Current Stock > 0	No recent sales — investigate	Red
SOR (6m) < 40% AND Age > 26 weeks	Underperforming — consider markdown or retire	Amber
WOC > 20 weeks AND SOR (6m) < 50%	Overstocked — transfer or markdown	Teal
Age > 8 weeks AND Reorder Count = 0 AND SOR (6m) < 50%	Tier 4 trial — review for retirement	Amber
Full Price % < 70%	Heavy discounting — review pricing strategy	Amber
SOR (6m) > 80% AND WOC 4–8 weeks	Strong performer — consider range extension	Green
All other cases	On review — monitor weekly	Blue

Data Refresh

The dashboard should refresh daily, pulling from the core styles table and the store-level sales table. A "Last Updated" timestamp should appear in the footer of every page. Where real-time data is available, WOC and ROS should update intraday.

Technical Notes for Builders

- The primary data source is a flat Excel/CSV file with one row per style and the columns listed in the Data Model section above.
- Store-level data is a second table joinable on Style Number.
- All KES values should be formatted with thousand separators (e.g. KES 4,615 or KES 13.4M for values over 1 million).
- Percentages should show one decimal place (e.g. 97.1%).
- WOC should show one decimal place (e.g. 1.6 wks).
- Trend indicators (up/down arrows with % change) should appear on every KPI card when comparison period data is available.
- The tool should be built as a web application (React + a charting library such as Recharts, Victory, or D2) or as a BI tool configuration (Power BI, Tableau, Metabase, or Looker Studio).
- For AI-assisted builds, the full data schema and sample data file should be provided alongside this prompt.